



SMART PHONE BATTERY MANAGEMENT SYSTEM

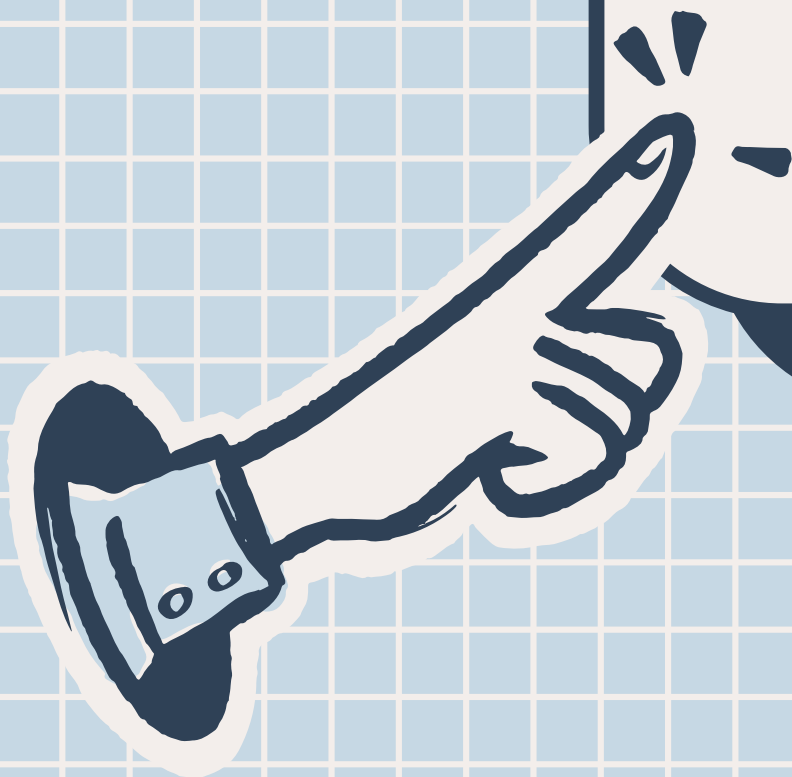
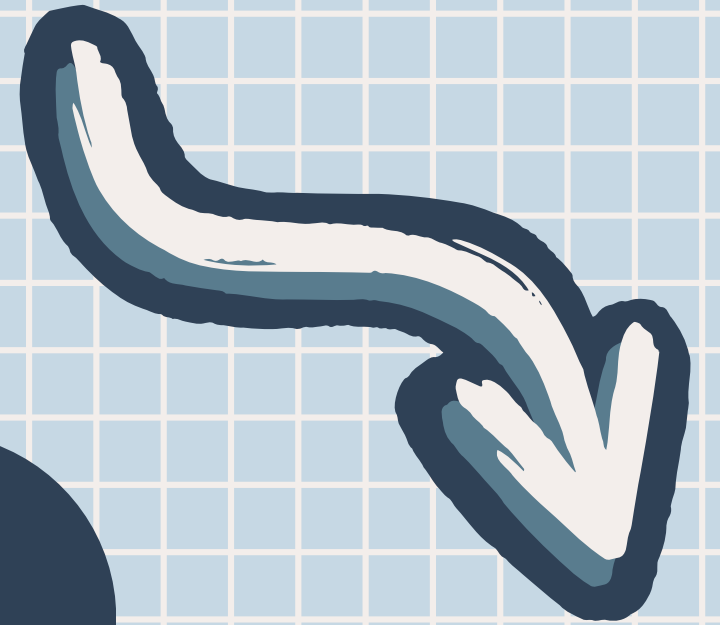
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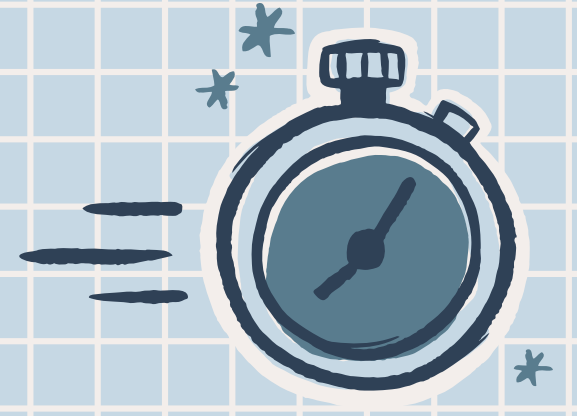
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ABSTRACT

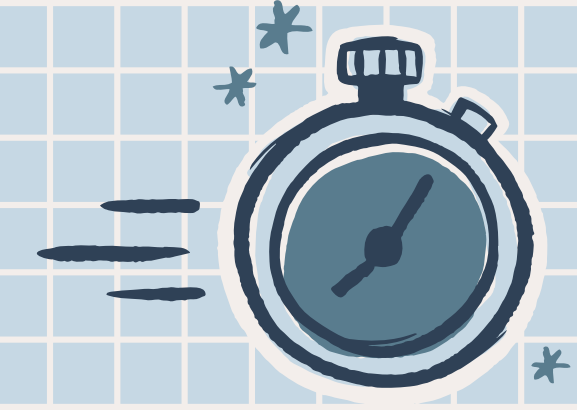


- Smartphones have become an essential part of daily life
- However, batteries drain quickly due to heavy usage and apps
- Traditional Battery Management Systems lack intelligent decision-making
- Proposed solution: Fuzzy Logic-Based Smart Battery Management System
- Implemented using Python and scikit-fuzzy library
- Uses Battery Level, Usage Type and Time as inputs
- Automatically selects Normal, Power Saver and Ultra Saver modes
- Improves battery life, safety and energy efficiency





INTRODUCTION

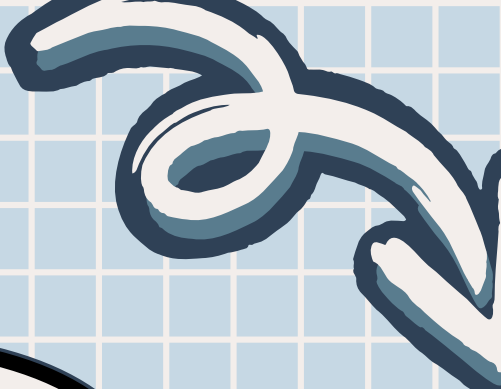


- ➔ Rapid growth of smartphones has increased the demand for efficient battery management
- ➔ High-performance apps and continuous connectivity lead to heavy power consumption
- ➔ Lithium-ion batteries face issues such as overheating, capacity fading & frequent charging
- ➔ Battery Management System ensures safe monitoring and control of battery operation
- ➔ Conventional BMS mainly performs basic monitoring and lacks intelligent decision-making
- ➔ Advanced and adaptive battery management techniques are required
- ➔ Hence, there is a need for a smart and intelligent battery management approach





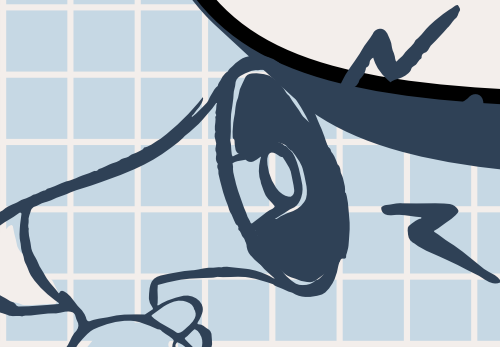
Problem Statement

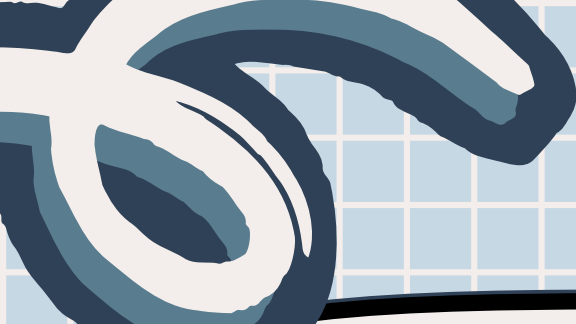


Smartphones make use of battery power based on user activities (example- gaming, studying, etc)

Modern battery management system do not explicitly integrate battery level, user activity and time of usage using intelligent decision-making techniques. This causes less efficient battery utilisation under varying usage conditions.

Therefore, there is a need for an intelligent battery management system that dynamically adjusts power-saving modes based on battery level, user activity, and time using fuzzy logic.





Objectives

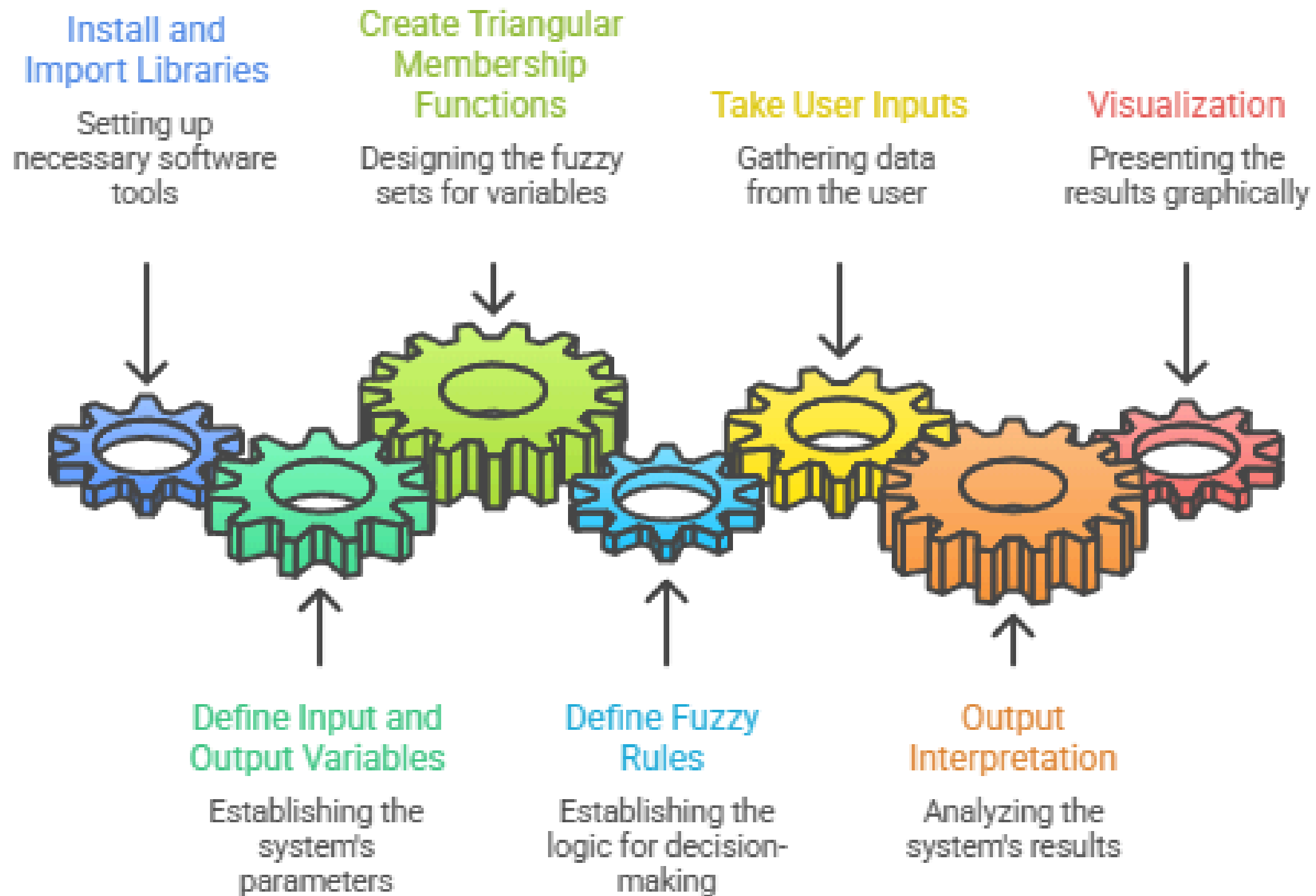


- To design a fuzzy logic based battery mode selection system using Python
- To define battery level, usage type, and time as input variables
- To create triangular membership functions for input and output variables
- To implement fuzzy IF-THEN rules for decision making
- To determine battery mode as Normal, Power Saver, or Ultra Saver
- To compute output using fuzzy inference and defuzzification
- To visualize membership functions of battery, usage time, and mode

RELATED WORK

- 1 Studies have explored data-driven methods for estimating battery health and predicting lifetime using machine learning.
- 2 Research on State of Health (SOH) prediction uses real-time data from electric vehicle platforms.
- 3 Machine learning models have been used to predict Remaining Useful Life (RUL) of batteries in applications like UAVs.
- 4 Thermal management models have been developed to control battery temperature and improve efficiency.
- 5 Other works focus on energy storage systems, cost optimization, and advanced battery materials.

Methodology



Methodology

1

Install and import libraries

The required libraries that are installed and imported are NumPy, Scikit-fuzzy and matplotlib

2

Define Input & Output Variables

Battery level , Usage type and time are input variables and mode is output variable that are defined

3

Triangular Membership functions are created

Battery (low, medium, high), Usage(Idle, study, gaming), Time(day & night) and Mode (normal, power, ultra) are triangular membership functions created

4

Define Fuzzy Rules

The following rules are implemented:

- ✿ If battery is low and usage is gaming → Ultra
- ✿ If battery is low and usage is study → Power
- ✿ If battery is medium and usage is gaming → Power
- ✿ If battery is high → Normal
- ✿ If battery is low and time is night → Ultra
- ✿ If battery is medium and usage is idle → Normal
- ✿ If battery is medium and time is day → Normal

All rules are combined using Control System and Control System Simulation

Methodology

5

Take user inputs

The user enters battery level, usage type and time. The system processes inputs using `sim.computer()` to generate output

6

Output Interpretation

The output is given as follows:

Less than 40 → Normal

40 to less than 70 → Power Saver

70 and above → Ultra Saver

Methodology

7

Visualisation

Membership plots are displayed using `battery.view()`, `usage.view()`, `time.view()`, `mode.view()` and matplotlib `plt.show()`

Workflow

```
graph LR; Input --> Fuzzification --> Rules --> Output
```

Input → Fuzzification → Rules → Output

LITERATURE SURVEY TABLE

YEAR	AUTHOR	PAPER TITLE	METHODOLOGY	FINDINGS	RELEVANCE
2024	Gopal Krishna, Rajesh Singh, Anita Gehlot, Ahmad Almogren, Ayman Altameem, Ateeq Ur Rehman, Seada Hussen	Advanced battery management system enhancement using IoT and ML.Focuses on predicting remaining useful life (RUL) of Li-ion batteries.	Uses IoT sensors with Arduino to collect real-time battery data.Applies LSTM machine learning model on NASA dataset for RUL prediction.	Proposed model achieved high accuracy with very low RMSE (0.01173). Outperformed existing models and effectively predicted battery life.	Improves battery life prediction and maintenance in EVs and energy systems. Supports efficient energy usage and sustainable development goals.
2024	Sambou Kone, Kumar Rahul Prasad, Faisal Saifi, Sagar Mishra, Sunil M P.	Smart Battery Management System for Portable Devices using Machine Learning. Focuses on improving battery life using ML techniques.	Uses sensors and ML models (ANN/GRNN) to monitor battery parameters.Predicts SOC, SOH, RUL and optimizes charging using data analysis.	ML model accurately predicts battery behavior and improves performance. System enhances battery life and prevents overheating/overcharging.	Useful for portable devices like laptops, power banks, and electronics. Improves efficiency, safety, and user experience in battery systems.

YEAR	AUTHOR	PAPER TITLE	METHODOLOGY	FINDINGS	RELEVANCE
2024	Daniel Flores-Martin, Sergio Laso, and Juan Luis Herrera.	Enhancing Smartphone Battery Life: A Deep Learning Model Based on User-Specific Application and Network Behavior.	The study utilized a custom Android app to collect real-time usage data (apps, network, battery) which was then used to train and compare Deep Neural Network (DNN) and Long Short-Term Memory (LSTM) models.	LSTM models proved superior for predicting energy drain by capturing temporal usage patterns, while the data collection process itself was found to have a negligible impact (0.07%) on total battery consumption.	This research enables personalized power management by predicting individual user behavior, offering a scalable way to extend smartphone battery life through proactive, data-driven resource optimization.
2024	Ms. Snehal Shinde and Dr. Nikita Kulkarni.	Issues and Challenges for Intelligent Battery Management System in Electric Vehicles.	Systematic review of 51 high-relevance articles selected from the Scopus database. Analyzed 25 specific studies to categorize technical and environmental BMS challenges.	Identified critical hurdles in real-time state estimation, data security, and thermal management. Notes that power source availability and disposal issues remain significant systemic barriers.	Provides a comprehensive overview of factors affecting EV efficiency and performance. Offers a foundational analysis for researchers to enhance intelligent battery monitoring systems
2020	Balasingam, B., Ahmed, M., and Pattipati, K.	Battery Management Systems—Challenges and Some Solutions	Review of model-based algorithms for fuel gauging, charging, and cell balancing. Focuses on equivalent circuit models (ECM) and weighted least squares estimation.	Effective BMS requires robust state-of-charge (SOC) and state-of-health (SOH) tracking. Key solutions include adaptive algorithms to manage battery aging and impedance	Critical for improving safety, efficiency, and reliability in electric vehicles. Provides a framework for universal BMS and secondary "second-life" battery use.

YEAR	AUTHOR	PAPER TITLE	METHODOLOGY	FINDINGS	RELEVANCE
2018	Pradeep M. Kumbar and Sowmya Sunkara. From BMS College of Engineering, Bengaluru.	Android Based Battery Monitoring System for Lithium Ion Batteries Used in EVs. Focuses on real-time monitoring using Android.	Uses sensors, Raspberry Pi, ADC, and web app to collect battery data. Data is sent to Android phone via cloud for real-time monitoring.	System successfully displays voltage, current, and temperature in real time. Prototype proves system is feasible and works effectively.	Improves safety and monitoring of EV batteries. Helps in remote maintenance and reduces failure risks.

